

Lecture VI - Minimizing Split Orders

Applied Optimization with Julia

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Introduction

E-Commerce Trends

Question: What are current trends in e-commerce?

E-Commerce Sales

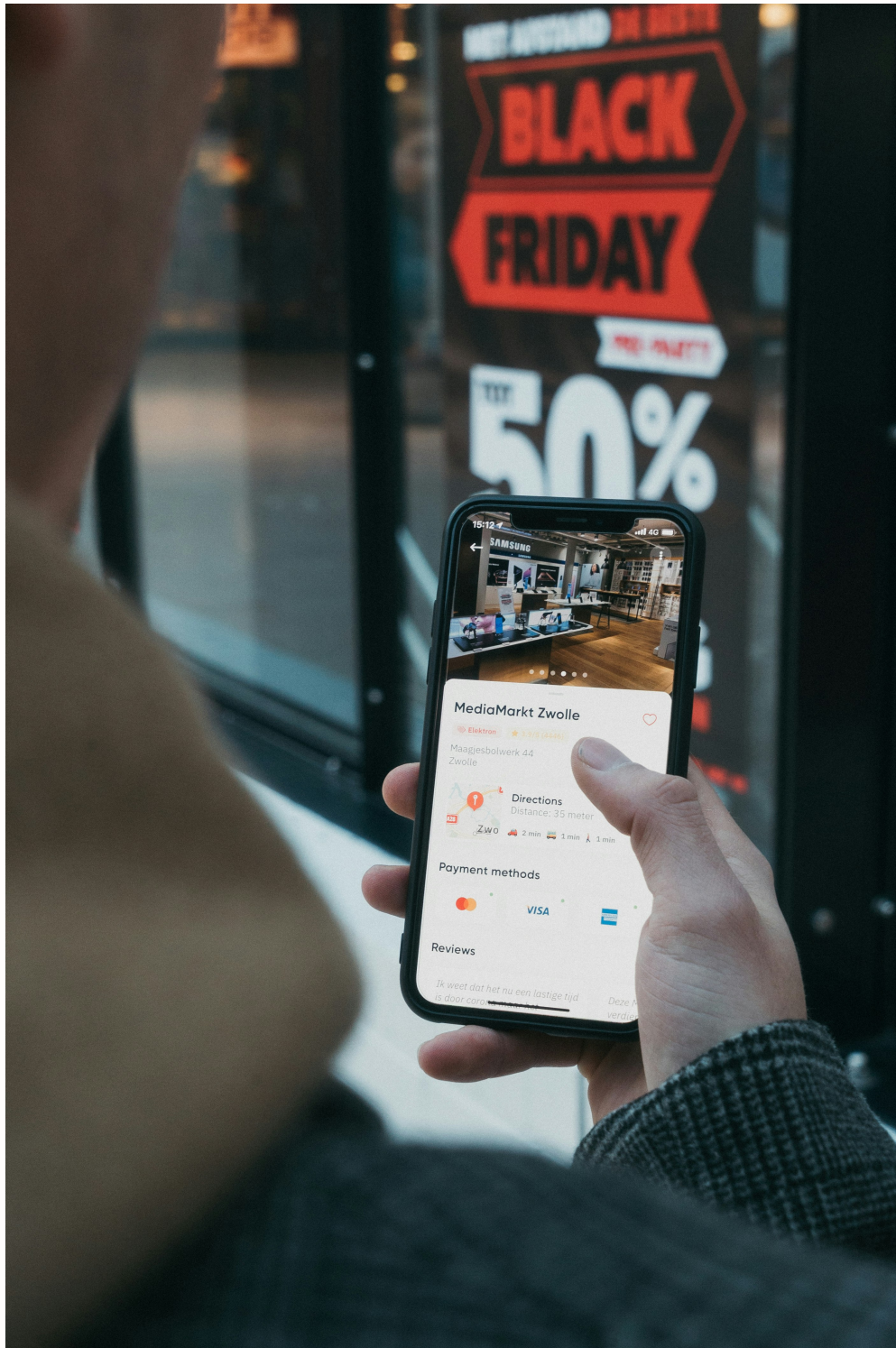
- E-Commerce sales are growing fast:
 - Products are **no longer bound between borders**
 - Product variety is **rising**
 - Consumer shopping patterns are **shifting**
 - Brick-and-mortar stores **lose customers to the internet**
 - Covid-19 **accelerated this trend even more**

Parcels Worldwide

- The number of parcels is rising:
 - **2014:** 44 billion parcels [1]
 - **2019:** 103 billion parcels [2]
 - **2026:** forecast of 220 – 262 billion¹ [3]

¹Forecast made in 2020, not an actual number

Pressure on infrastructure



- Consumers nowadays expect free, fast deliveries and returns
- Existing warehouses have to store an **increasing range of products**
- Better customer service requires **faster deliveries**
- Incurred fulfillment costs **depend on the number of parcels**

Pressure on the environment



- **Each parcel packaging** consumes resources during production
- Every dispatched parcel to the customer causes CO₂ emissions
- In case of returns, **more parcels cause more emissions**

Learning Objectives

After this lecture, you will be able to:

- **Explain** split orders and why they matter in e-commerce
- **Build** a coappearance matrix from transactional data
- **Formulate and solve** the Quadratic Multiple Knapsack Problem (QMKP) in JuMP

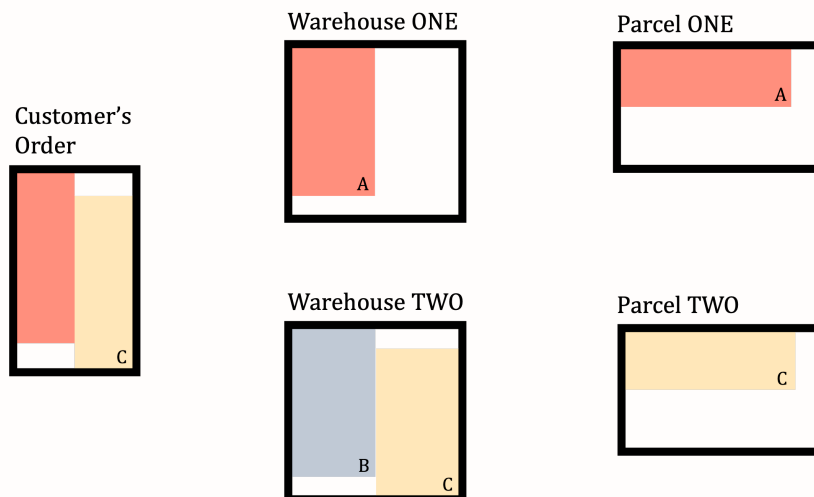
- **Judge** when exact solvers reach their limits and heuristics are needed

Problem Structure

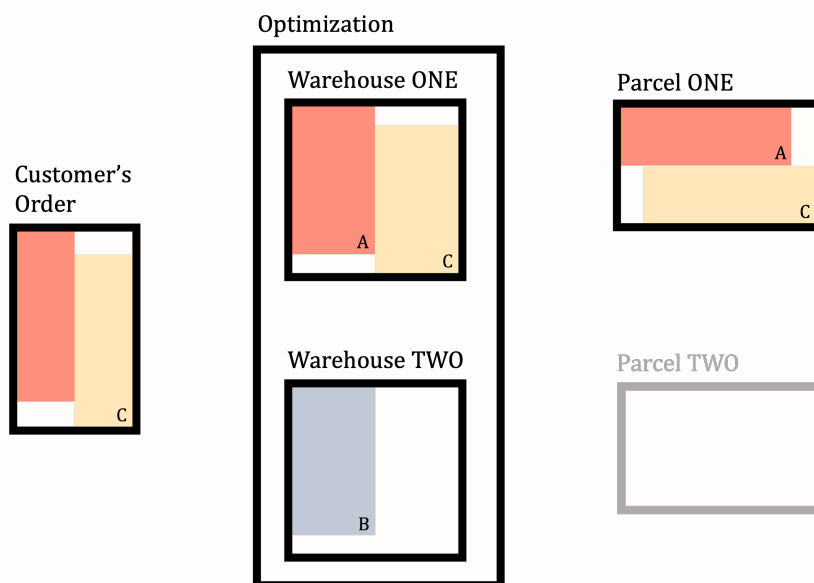
Split Order

Question: **What is a split order?**

...



No Split Order



Reason for Split Orders

Question: **Why might they occur?**

...

- **Stock availability:** Some products are out of stock at a warehouse and need to be fulfilled from another warehouse

- **Capacity constraints:** Some products are stored at different warehouses and need to be shipped from elsewhere

Impact of Split Orders

Question: **What are the consequences?**

...

- **Higher** shipping costs
- **Increased** packaging material
- **More** CO₂ emissions
- Higher operational complexity
- **Lower** customer satisfaction

Mitigations?

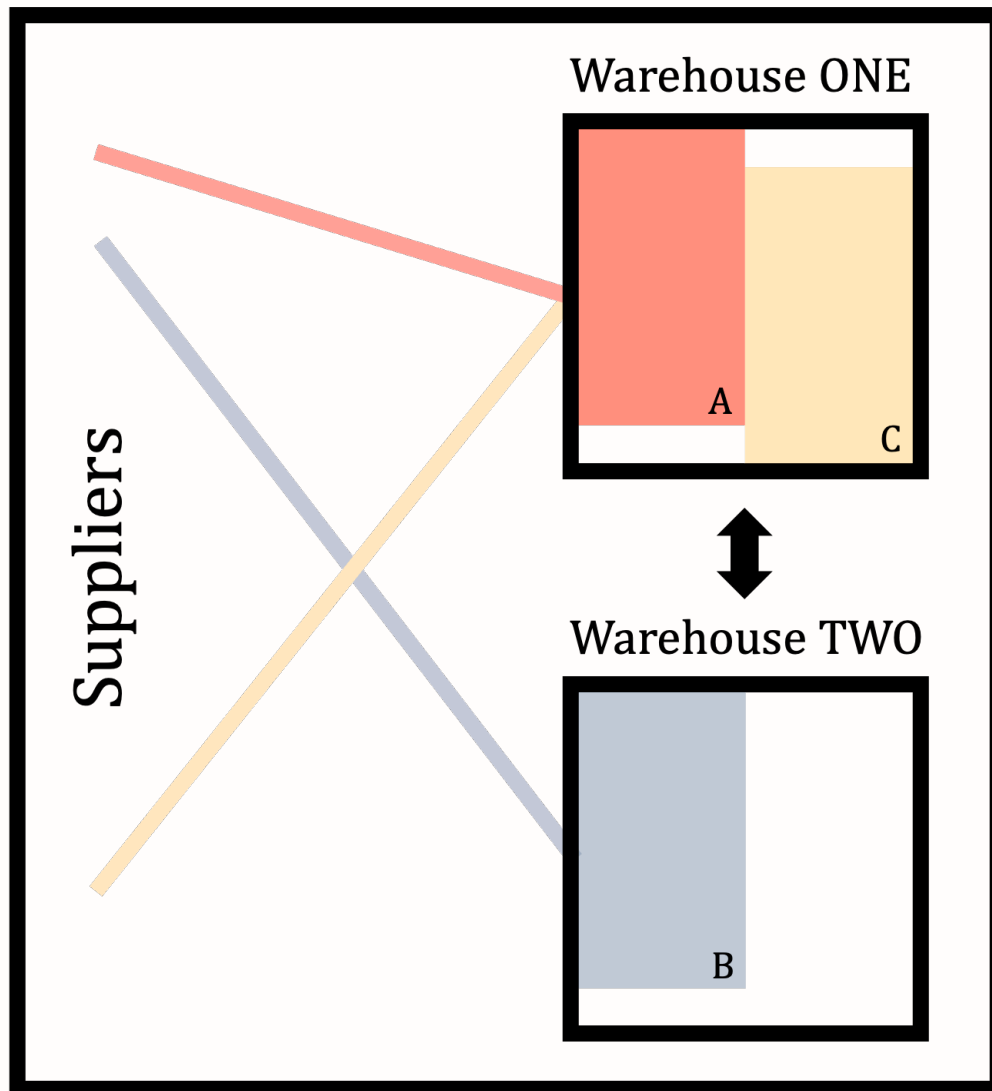
Question: **What are possible mitigations?**

...

- **Consolidation:** Ship to a central warehouse before dispatch
- **Cross-docking:** Transfer goods from inbound to outbound transport with little or no storage in between
- **Transshipment:** Ship between warehouses before delivery
- **Co-allocation:** Predict coappearance of products and allocate them to the same warehouse

Problem Structure - Version 1

Optimizing Co-allocation



Question: What could be our objective?

We aim to improve the **SKU²-warehouse allocation** to minimize the number of split parcels resulting from **SKUs being stored in different warehouses**.

Available Sets

Question: **What could be the sets here?**

- $\mathcal{I}$ - Set of products indexed by $i \in \{1, 2, \dots, |\mathcal{I}|\}$
- $\mathcal{K}$ - Set of warehouses indexed by $k \in \{1, \dots, |\mathcal{K}|\}$
- $\mathcal{M}$ - Set of customer orders $m \in \{1, 2, \dots, |\mathcal{M}|\}$

²SKU: Stock Keeping Unit

Available Parameters

Question: **What are possible parameters?**

- c_k - Storage space of warehouse $k \in \{1, \dots, |\mathcal{K}|\}$, measured in number of SKUs
- $T = (t_{m,i})$ - Past customer orders for SKUs

...

Question: **What could the transactional data look like?**

Transactional Data

$t_{m,i}$	A	B	C	D
1	1	1	1	0
2	1	1	1	0
3	1	1	0	0
4	1	0	0	1
5	1	0	0	1
6	1	0	0	1
7	1	0	0	1
8	0	0	1	1

Past vs. Future

- The transactional data T is based on **past orders**
- It is a **binary matrix** of customer orders and SKUs
- We use this data to **assume** future co-occurrence
 - Past co-occurrence predicts future co-occurrence

...

Question: **What is your opinion on the assumption?**

Split-Order Minimization

Question: **What could be our decision variable/s?**

...

i We have the following sets:

- $\mathcal{I}$ - Set of products indexed by $i \in \{1, 2, \dots, |\mathcal{I}|\}$
- $\mathcal{K}$ - Set of warehouses indexed by $k \in \{1, \dots, |\mathcal{K}|\}$
- $\mathcal{M}$ - Set of customer orders $m \in \{1, 2, \dots, |\mathcal{M}|\}$

...

- $X_{i,k}$ - 1, if $i \in \mathcal{I}$ is stored in $k \in \mathcal{K}$, 0 otherwise

- $Y_{m,i,k} = 1$, if SKU $i \in \mathcal{I}$ is shipped from warehouse $k \in \mathcal{K}$ for customer order $m \in \mathcal{M}$,
0 otherwise

Integer Programming Model

- A. Catalán and M. Fisher [4] created an **integer model**
- Number of SKUs of E-Commerce retailers can easily be **between 10,000 - 100,000**
- The number of customer orders needed for “stable” results is one order of magnitude higher: **100,000 - 10,000,000**

...

Question: **Does anyone have an idea what this could mean?**

Why is Version 1 so hard?

- Objective: **minimize the number of warehouses** that ship parcels for each customer order
- Requires $Y_{m,i,k}$: one variable per order, SKU **and** warehouse
- $\rightarrow |\mathcal{M}| \times |\mathcal{I}| \times |\mathcal{K}|$ binary variables

...

! Variables explode with the instance size

10,000 SKUs, 1,000,000 orders and 2 warehouses already lead to 2×10^{10} binary variables $Y_{m,i,k}$!

Implementation Challenges

- Small instance with 10 SKUs and 1000 customer orders
- **CPLEX 20.1.0** needs 3100 seconds to solve the problem
- Computation time scales exponentially
- $\rightarrow$ **Not applicable** in real world applications!

Any idea what

could be done?

Problem Structure - Version 2

Heuristic Approach

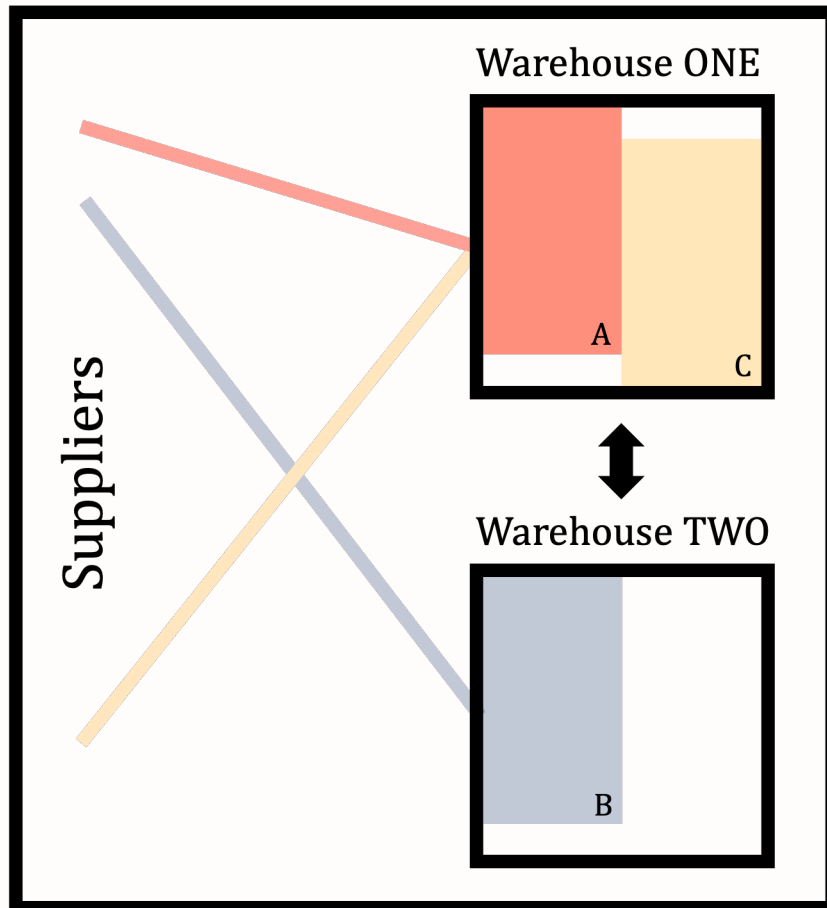
- **Heuristic**: Fast, but not necessarily optimal
- **Approximation**: Not guaranteed to be optimal, but close
- **Computational Effort**: Reasonable even for large instances

...

💡 Different view on the problem

Focus on the warehouses and the coappearance of SKUs! Discard the exact information about the customer orders.

Objective



Question: **What could be the objective?**

Maximize the coappearance of products that are often **part of the same customer orders**.

Transaction Matrix

```
T = [  
  1 1 1 0;  
  1 1 1 0;  
  1 1 0 0;  
  1 0 0 1;  
  1 0 0 1;  
  1 0 0 1;  
  1 0 0 1;  
  1 0 0 1;
```

```

    0 0 1 1
]

# Create the coappearance matrix
Q = T' * T
println("Coappearance matrix Q:")
display(Q)

```

Coappearance matrix Q:

```

4x4 Matrix{Int64}:
 7  3  2  4
 3  3  2  0
 2  2  3  1
 4  0  1  5

```

Coappearance Matrix

- Q is a symmetric matrix
- Proposed by A. Catalán and M. Fisher [4]
- $Q = T^T T$ where $Q = (q_{i,j})_{i,j \in \{1, \dots, |\mathcal{I}|\}}$
- $q_{i,j}$ shows how often i and j appear **in the same order**

...

Question: **What do the principal diagonal values tell us?**

...

- In how many **orders** each SKU appeared
- As $t_{m,i}$ is binary, $q_{i,i}$ counts orders, **not units sold**

How to approach the problem?

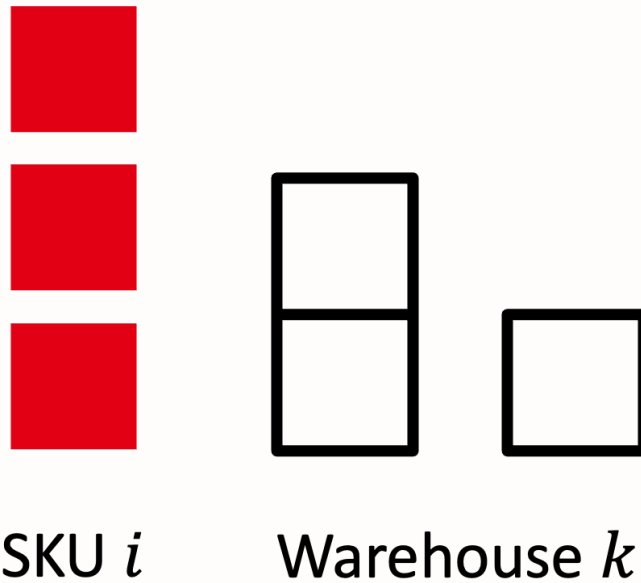
- **Greedy Heuristic**³: Allocation based on matrix
- **Mathematical Model**⁴: Maximizes coappearance
- **GRASP**⁵: Good on small instances
- New: Max. coappearance with non-linear solver
- New: Heuristic based on Chi-Square Tests

³Simple and very fast, A. Catalán and M. Fisher [4]

⁴Computationally intensive with CPLEX, S. Zhu, X. Hu, K. Huang, and Y. Yuan [5]

⁵Greedy Randomized Adaptive Search Procedure, S. Zhu, X. Hu, K. Huang, and Y. Yuan [5]

Basic Setting



Available Data (Version 2)

Question: **What could be the sets?**

...

- $\mathcal{I}$ - Set of products indexed by $i \in \{1, 2, \dots, |\mathcal{I}|\}$
- $\mathcal{K}$ - Set of warehouses indexed by $k \in \{1, \dots, |\mathcal{K}|\}$

...

! No customer order information is needed!

We can focus on the SKUs and the warehouses, making the problem **much smaller!**

Available Parameters

Question: **What are possible parameters?**

- c_k - Storage space of warehouse $k \in \{1, \dots, |\mathcal{K}|\}$, measured in number of SKUs
- $Q = (q_{i,j})_{i,j \in \{1, \dots, |\mathcal{I}|\}}$ - Coappearance matrix

...

! Transactional Data replaced

Instead of the transactional data, we just **use the coappearance matrix** in our model!

Model Formulation

Decision Variables?

i We have the following sets:

- $\mathcal{I}$ - Set of products indexed by $i \in \{1, 2, \dots, |\mathcal{I}|\}$
- $\mathcal{K}$ - Set of warehouses indexed by $k \in \{1, \dots, |\mathcal{K}|\}$

...

! Our objective is to:

Maximize the coappearance of products that are often part of the same customer orders. **In more mathematical terms:** Maximize the sum of all unique pair-wise values $q_{i,j}$ of all SKUs stored in the same warehouse.

...

Question: **What could be our decision variable/s?**

Decision Variables

- $X_{i,k}$ - 1, if SKU $i \in \mathcal{I}$ is stored in $k \in \mathcal{K}$, 0 otherwise

...

! Only one variable per SKU and warehouse!

As we don't need the customer order information, we only need to make a decision for each SKU and warehouse pair!

Decision Variable in Julia

Question: **How could we formulate the variable in Julia?**

```
using JuMP, SCIP # SCIP is a non-commercial solver for quadratic models

warehouses = ["Hamburg", "Berlin"] # Add warehouses as a vector
skus = ["A", "B", "C", "D"] # The four SKUs from our transactional data

warehouse_model = Model(SCIP.Optimizer)
```

...

```
@variable(warehouse_model, X[i in skus, k in warehouses], Bin)
```

```
2-dimensional DenseAxisArray{JuMP.VariableRef,2,...} with index sets:
  Dimension 1, ["A", "B", "C", "D"]
  Dimension 2, ["Hamburg", "Berlin"]
And data, a 4x2 Matrix{JuMP.VariableRef}:
X[A,Hamburg] X[A,Berlin]
X[B,Hamburg] X[B,Berlin]
X[C,Hamburg] X[C,Berlin]
X[D,Hamburg] X[D,Berlin]
```

Objective Function

i We need the following:

- $X_{i,k}$ - 1, if SKU $i \in \mathcal{I}$ is stored in $k \in \mathcal{K}$, 0 otherwise
- $q_{i,j}$ - Coappearance of SKU $i \in \mathcal{I}$ and $j \in \mathcal{I}$

! Our objective is to:

Maximize the sum of all unique pair-wise values $q_{i,j}$ of all SKUs stored in the same warehouse. Note, that this is a **quadratic objective function**!

...

Question: **What could the objective function look like?**

...

Quadratic Objective Function

$$\text{Maximize} \quad \sum_{i=2}^{|\mathcal{I}|} \sum_{j=1}^{i-1} \sum_{k \in \mathcal{K}} X_{i,k} \times X_{j,k} \times q_{i,j}$$

...

i This is a **quadratic objective function**!

The quadratic terms are $X_{i,k} \times X_{j,k}$. This objective function is based on the **Quadratic Multiple Knapsack Problem (QMKP)**, formulated by A. Hiley and B. A. Julstrom [6].

Objective Function in Julia

Question: **How could we formulate this in Julia?**

...

```
# We reuse the matrix Q = T' * T computed from our transactional data
```

```

@objective(warehouse_model,
    Max,
    sum(
        X[skus[i], warehouses[k]] * X[skus[j], warehouses[k]] * Q[i,j]
        for i in 2:length(skus)
        for j in 1:i-1
        for k in 1:length(warehouses)
    )
)

```

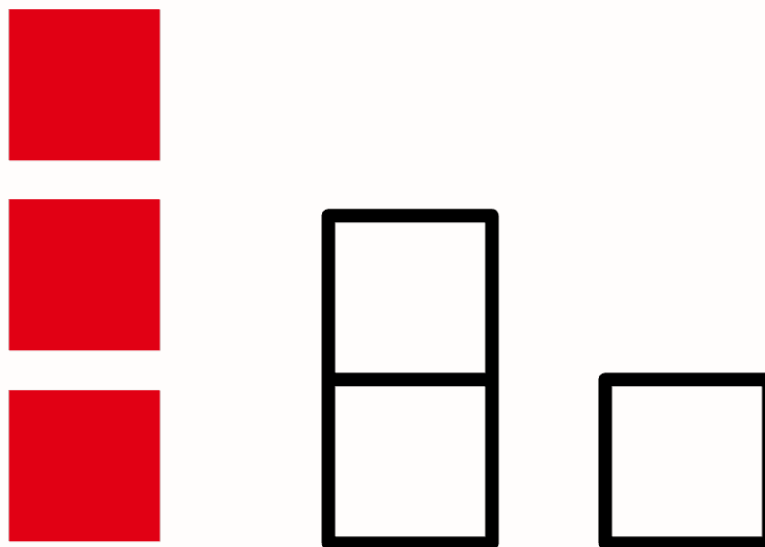
```

3      X[B,Hamburg]*X[A,Hamburg]      +      3      X[B,Berlin]*X[A,Berlin]      +
2      X[C,Hamburg]*X[A,Hamburg]      +      2      X[C,Berlin]*X[A,Berlin]      +
2      X[C,Hamburg]*X[B,Hamburg]      +      2      X[C,Berlin]*X[B,Berlin]      +
4      X[D,Hamburg]*X[A,Hamburg]      +      4      X[D,Berlin]*X[A,Berlin]      +
X[D,Hamburg]*X[C,Hamburg] + X[D,Berlin]*X[C,Berlin]

```

Constraints

What constraints?



SKU i

Warehouse k

Question: **What constraints?**

- Allocate each SKU **at least once**
- Warehouses have a **finite capacity**
- Capacity is **not exceeded**

Single Allocation Constraint?

! The goal of this constraint is to:

Ensure that each SKU is allocated at least once.

...

i We need the following variable:

- $X_{i,k}$ - 1, if SKU $i \in \mathcal{I}$ is stored in $k \in \mathcal{K}$, 0 otherwise

...

Question: What could the constraint look like?

Single Allocation Constraint

$$\sum_{k \in \mathcal{K}} X_{i,k} \geq 1 \quad \forall i \in \mathcal{I}$$

...

i Remember, this is the variable:

- $X_{i,k}$ - 1, if SKU $i \in \mathcal{I}$ is stored in $k \in \mathcal{K}$, 0 otherwise

...

Question: How could we change the constraint to ensure that each SKU is allocated only once?

...

Question: How could we add the constraint in Julia?

Single Allocation in Julia

```
@constraint(warehouse_model, single_allocation[i in skus],
    sum(X[i, k] for k in warehouses) >= 1
)
```

1-dimensional DenseAxisArray{JuMP.ConstraintRef{JuMP.Model, MathOptInterface.ConstraintIndex{MathOptInterface.GreaterThan{Float64}}}, JuMP.ScalarShape{1,...}} with index sets:

Dimension 1, ["A", "B", "C", "D"]

And data, a 4-element Vector{JuMP.ConstraintRef{JuMP.Model, MathOptInterface.ConstraintIndex{MathOptInterface.GreaterThan{Float64}}}, JuMP.ScalarShape{1,...}}:

```
single_allocation[A] : X[A,Hamburg] + X[A,Berlin] ≥ 1
single_allocation[B] : X[B,Hamburg] + X[B,Berlin] ≥ 1
```

```
single_allocation[C] : X[C,Hamburg] + X[C,Berlin] ≥ 1
single_allocation[D] : X[D,Hamburg] + X[D,Berlin] ≥ 1
```

Capacity Constraints?

! The goal of these constraints is to:

Ensure that the capacity of each warehouse is not exceeded.

...

i We need the following variables and parameters:

- $X_{i,k}$ - 1, if SKU $i \in \mathcal{I}$ is stored in $k \in \mathcal{K}$, 0 otherwise
- c_k - Storage space of warehouse $k \in \mathcal{K}$

...

Question: **What could the second constraint be?**

Capacity Constraints

$$\sum_{i \in \mathcal{I}} X_{i,k} \leq c_k \quad \forall k \in \mathcal{K}$$

...

And that's basically it!

...

Question: **How could we add the second constraint in Julia?**

Capacity Constraints in Julia

```
capacities = Dict{"Hamburg" => 2, "Berlin" => 2} # Add capacities
```

```
@constraint(warehouse_model, capacity[k in warehouses],
    sum(X[i, k] for i in skus) <= capacities[k]
)
```

1-dimensional DenseAxisArray{JuMP.ConstraintRef{JuMP.Model, MathOptInterface.ConstraintIndex{MathOptInterface.LessThan{Float64}}}, JuMP.ScalarShape{1,...}} with index sets:

```
Dimension 1, ["Hamburg", "Berlin"]
```

And data, a 2-element Vector{JuMP.ConstraintRef{JuMP.Model, MathOptInterface.ConstraintIndex{MathOptInterface.LessThan{Float64}}}, JuMP.ScalarShape{1,...}}:

```
capacity[Hamburg] : X[A,Hamburg] + X[B,Hamburg] + X[C,Hamburg] + X[D,Hamburg]
≤ 2
capacity[Berlin] : X[A,Berlin] + X[B,Berlin] + X[C,Berlin] + X[D,Berlin] ≤ 2
```

QMK Model

$$\text{Maximize} \quad \sum_{i=2}^{|\mathcal{I}|} \sum_{j=1}^{i-1} \sum_{k \in \mathcal{K}} X_{i,k} \times X_{j,k} \times q_{i,j}$$

subject to:

$$\begin{aligned} \sum_{k \in \mathcal{K}} X_{i,k} &\geq 1 \quad \forall i \in \mathcal{I} \\ \sum_{i \in \mathcal{I}} X_{i,k} &\leq c_k \quad \forall k \in \mathcal{K} \\ X_{i,k} &\in \{0, 1\} \quad \forall i \in \mathcal{I}, k \in \mathcal{K} \end{aligned}$$

QMK Model in Julia

```
set_attribute(warehouse_model, "display/verblevel", 0) # Hide solver output
optimize!(warehouse_model)
```

```
println("The optimal objective value is: ", objective_value(warehouse_model))
println("The optimal solution is: ", value.(X))
```

```
The optimal objective value is: 6.0
The optimal solution is: 2-dimensional DenseAxisArray{Float64,2,...} with
index sets:
  Dimension 1, ["A", "B", "C", "D"]
  Dimension 2, ["Hamburg", "Berlin"]
And data, a 4x2 Matrix{Float64}:
 1.0  0.0
 0.0  1.0
 0.0  1.0
 1.0  0.0
```

...

Question: **What does this value tell us?**

Interpreting the Solution

- Total capacity (2 + 2) **equals** the number of SKUs
- Hence, each SKU is stored **exactly once**
- Optimal: **A and D** share a warehouse, so do **B and C**

...

i The objective value is 6

It sums the coappearances of SKUs sharing a warehouse: $q_{A,D} + q_{B,C} = 4 + 2 = 6$.
It does **not** directly count the avoided split orders!

Model Characteristics

Characteristics

- Is the model formulation **linear/ non-linear**?
- What kind of **variable domain** do we have?
- Do we know the **split-orders** based on the **objective value**?
- Why **couldn't we use HiGHS** as solver?

Choosing a solver

- Identify **problem structure**, e.g. LP, MIP, NLP, MIQP, MIQCP, ...
- What is the **size** of the problem?
- Is a **commercial** solver needed?

...

i Commercial Solvers

Commercial solvers are **faster** and **more robust** than open source solvers but also **more expensive**. During your studies, you can use most of them for free though! Nonetheless, we will only use open source solvers in this course.

Our Problem Class

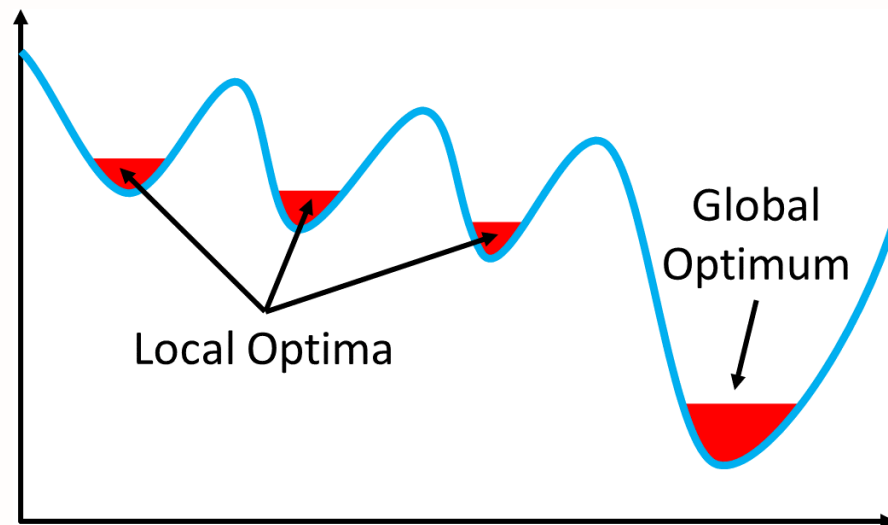
- Objective: **quadratic** terms $X_{i,k} \times X_{j,k}$
- Constraints: all **linear**
- Variables: all **binary**

...

i Mixed-Integer Quadratic Program (MIQP)

Our model is a **MIQP**, as only the objective is quadratic. It is not a MIQCP, which would also allow quadratic **constraints**. MIQCP solvers, e.g. SCIP, subsume MIQPs and can thus solve our model.

Global vs Local Optimality



Solver Comparison⁶

SKUs	SCIP	HiGHS	Gurobi
100	118s	X	~400s
1,000	1,011s (18%)	X	~200s (2%)
10,000	X	X	X

Solver Comparison: Takeaways

- SCIP: Best open-source for MIQP/MIQCP, limited scalability
- **Commercial solvers:** Better but still fail on 10,000+ SKUs
- **Conclusion:** Heuristics necessary for realistic problems!

Model Assumptions

Questions: **On model assumptions**

- What assumptions have we made?
- Problem with allocating SKUs to multiple warehouses?
- What else might pose a problem in the real world?

...

i Duplicates can inflate the objective

With slack capacity, the model may store one SKU in **several warehouses**. A shared pair then counts **once per warehouse**, raising the objective without avoiding any further splits.

⁶Based on numerical experiments with QMK instances. Times are averages over the instances solved to optimality; the percentage shows the share of instances solved within one hour, X = none solved within one hour.

Model Limitations

- **Stock-outs:** Model assumes perfect availability
 - Real stockouts create splits despite optimal allocation
- **Storage costs:** No differentiation between costs
- **Demand dynamics:** Weather, promotions, seasonality
- **SKU sizes:** Uniform capacity assumption unrealistic
- **Shipping costs:** No consideration of distance or weight

Practical Limitations

Allocation changes:

- Frequent reallocation disrupts operations
- Physical inventory movements costly
- Need for change minimization

Workload imbalance:

- Optimal allocation may overload some warehouses
- High-frequency SKUs cluster together
- Trade-off: splits vs. balance

Impact

Can this be
applied?

Problem Size is Crucial

- Up to 1,000 SKUs → **commercial solvers**
- More than 1,000 SKUs → **heuristics**
- For example, the CHI heuristic

...

i CHI-Heuristic

Detect dependencies between products and allocate them accordingly, as products within orders can have dependencies and products are bought with different frequencies!

CHI Heuristic

Statistical Independence

Question: **When are two SKUs independent?**

...

If purchasing one doesn't affect purchasing the other:

$$P(A \cap B) = P(A) \times P(B)$$

...

Example: If A appears in 10% of orders and B in 20%:

- **Independent:** Appear together in $0.10 \times 0.20 = 2\%$ of orders
- **Dependent:** Appear together in 5% of orders → customers buy them together!

Chi-Square Test Intuition

Use **chi-square tests** to detect if SKUs are dependent:

- Calculate expected coappearances under independence
- Compare with **actual coappearances**
- If difference is significant → SKUs are **dependent**
- Allocate dependent SKUs to same warehouse!

From Tests to Allocations

💡 Two-Phase Approach

Phase 1: Build allocation using dependencies

Phase 2: Refine with local search

...

Result: Works on large instances, 50,000 SKUs in 10 minutes!

Practical Implementation

Real-World Challenges

Question: **What challenges might arise in practice?**

- **New products:** Allocate SKUs without historical data?
- **Dynamic inventory:** SKUs appear and disappear
- **Different sizes:** SKUs consume varying storage space
- **Computational limits:** Problems with 50,000+ SKUs

Rolling Horizon Approach

In the following case study [7], we:

- used a **5-week training window** for allocation decisions
- updated allocations **weekly** based on recent patterns
- balanced stability vs. responsiveness

...

💡 Tip

In the case study, SKUs were grouped by **category-brand combinations** to reduce the problem size while maintaining allocation quality. New SKUs inherit the allocation of their cluster!

Case Study

Problem characteristics:

- 100,000+ SKUs
- Several million orders
- Multiple warehouses
- Storage capacity constraints

Heuristics compared⁷:

- **CHI**: Chi-Square tests [7]
- **GP, GO, GS, BS**: Greedy [4]
- **RA**: Random allocation

Implementation Results

Retailer's status quo: 6.95% of all orders were split

...

Theoretical reduction of these splits:

- CHI: **82.25%** reduction
- BS: **62.63%** reduction
- GS: **59.16%** reduction

Conclusion

- Splits are **of no benefit**, except **faster customer deliveries**
- Increase workload, packaging and shipping costs
- Mathematical Optimization of “**full**” **problem not solvable**
- **CHI** Heuristic close to mathematical optimization

...

i And that's it for today's lecture!

We now have covered the Quadratic Multiple Knapsack Problem and are ready to start solving some tasks in the upcoming tutorial.

Questions?

⁷QMKP is not applicable for instance in case study

Literature

Literature I

References for this lecture:

- A. Catalán and M. Fisher [4]: Integer model and greedy heuristics
- A. Hiley and B. A. Julstrom [6]: The Quadratic Multiple Knapsack Problem
- S. Zhu, X. Hu, K. Huang, and Y. Yuan [5]: Mathematical model and GRASP
- T. Vlček and G. Voigt [7]: CHI heuristic and case study

Literature II

For more interesting literature to learn more about Julia, take a look at the [literature list](#) of this course.

Bibliography

- [1] Pitney Bowes Inc., “Pitney Bowes Parcel Shipping Index Reveals 48 Percent Growth in Parcel Volume since 2014.” Accessed: May 27, 2021. [Online]. Available: <https://www.businesswire.com/news/home/20170830005628/en/Pitney-Bowes-Parcel-Shipping-Index-Reveals-48>
- [2] Pitney Bowes Inc., “Pitney Bowes Parcel Shipping Index Reports Continued Growth Bolstered by China and Emerging Markets.” Accessed: May 27, 2021. [Online]. Available: <https://www.businesswire.com/news/home/20191010005148/en/>
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